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Harnessing Biomedical Literature to Calibrate Clinicians' Trust in AI Decision Support Systems

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ABSTRACT

Clinical decision support tools (DSTs), powered by Artificial Intelligence (AI), promise to improve clinicians' diagnostic and treatment decision-making. However, no AI model is always correct. DSTs must enable clinicians to validate each AI suggestion, convincing them to take the correct suggestions while rejecting its errors. While prior work often tried to do so by explaining AI's inner workings or performance, we chose a different approach: **We investigated how clinicians validated each other's suggestions in practice (often by referencing scientific literature) and designed a new DST that embraces these naturalistic interactions. This design uses GPT-3 to draw literature evidence that shows the AI suggestions' robustness and applicability (or the lack thereof).** A prototyping study with clinicians from three disease areas proved this approach promising. Clinicians' interactions with the prototype also revealed new design and research opportunities around (1) harnessing the complementary strengths of literature-based and predictive decision supports; (2) mitigating risks of de-skilling clinicians; and (3) offering low-data decision support with literature.

CCS CONCEPTS

• **Information systems** → Information extraction; • **Applied computing** → **Health care information systems**; • **Human-centered computing** → **Empirical studies in HCI**.

KEYWORDS

Clinical AI, XAI, Biomedical Literature, Qualitative Method

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1 INTRODUCTION

The idea of leveraging machine intelligence to improve clinical decision-making has fascinated healthcare and Artificial Intelligence (AI) researchers for decades. Today, diverse AI systems have proven their performance in research labs and are moving into clinical practice in the form of Decision Support Systems (DSTs). From Bayesian models that predict treatment outcomes based on Electronic Health Records (EHR) [67] to computer vision systems that interpret medical images [39], from rule-based systems that alert drug interactions [10] to wearable-sensing AIs that monitor disease progression [18, 56], **AI-powered DSTs promise to reduce clinician decision errors and improve patient outcomes.**

While all clinical AI models can offer valuable diagnostic or treatment suggestions, none is always correct. **Therefore, clinicians must calibrate their trust in each AI suggestion on a case-by-case basis.** DST interaction designs can help. Ideally, clinical DSTs can provide information that enables clinicians to adopt only the correct AI suggestions while staying unbiased by AI errors.

This is a challenging goal. Amongst other approaches, **existing DST designs most often supported clinician-AI trust calibration by explaining how the AI generated its suggestions and how well it performed on past patient data [26, 35, 54, 73]. This "explanation" approach struggled in clinical practice: When the clinician's hypothesis was wrong and the AI advice was correct, the explanations rarely persuaded clinicians to take the advice [14, 26, 32, 36, 41, 74]. When the AI suggestion was wrong, explanations could also fail to help clinicians notice the error [21, 73, 75].** These failures propelled some researchers to design more persuasive AI explanations, for example, by presenting only parts of the explanations that justify the AI's output [17]. However, these designs are emergent, and they have not yet empirically demonstrated their effectiveness.

This paper aims to identify new DST designs that can effectively calibrate clinicians' trust in AI suggestions on a case-by-case basis, enabling them to take only correct suggestions while rejecting its errors. Instead of exploring new ways to explain the AI and testing them on clinicians afterward, we chose to **start by investigating how**

clinicians validated each other's diagnostic or treatment suggestions in practice. We hoped that these naturalistic interactions would reveal new insights into the kinds of information clinicians need to calibrate their trust in AI suggestions. Given that biomedical literature (e.g., reports of randomized controlled trials) is known to play a central role in this process [31] and that clinicians had rejected AI suggestions because “they have not been published in prestigious clinical journals” [25, 75], we paid particular attention to how clinicians sought and used evidence from the literature.

Through contextual inquiries with 12 clinicians and their assistants, we found that clinicians rarely explained how they came up with the suggestion when exchanging diagnostic or treatment suggestions. Instead, they sought all evidence that may validate or invalidate the suggestion from biomedical literature, their shared source of truth. They then examined these pieces of evidence based on the evidence's applicability to the specific patient situation at hand, thereby concluding whether to accept the suggestion.

Embracing these findings, we designed a novel form of DST that imitates clinicians' natural trust-calibration interactions. Rather than explaining how the AI generated its suggestion, the DST provides literature evidence that can potentially validate or invalidate the suggestions. In presenting the evidence, the DST highlights its applicability to the particular patient situation in question, rather than how rigorously it was concluded from past patient cases. A co-prototyping study with 9 clinicians confirmed these design strategies. Further, clinicians' interactions with the prototype surfaced additional design and research opportunities around designing more intuitive clinician-AI trust-calibration interactions and harnessing biomedical literature for creating such interactions.

This paper presents the initial contextual inquiries, the new DST design, and findings from the prototyping study. It makes three contributions. First, this work provides a rare description of how clinicians sought information to validate each other's diagnostic/treatment suggestions in their practice. It provides a timely answer to researchers' call for a use-context-focused approach to designing explainable AI [43]. Second, this work identifies an alternative approach to calibrating clinicians' trust in AI. Our prototype exemplifies one possibility in this new design space. Third, the prototyping study offers an initial description of how clinicians deliberate clinical decisions using a predictive DST and biomedical literature simultaneously. Both clinical DST and Biomedical Natural Language Processing (biomedNLP) technologies are rapidly maturing. This research offers a valuable reference for future research that harness both technologies to improve clinical decision-making.

2 RELATED WORK

2.1 Challenges of Validating AI Suggestions with “Explainable AI”

Across the fields of HCI and AI, extensive research has studied how to calibrate clinicians' trust in AI. Much of this research can trace its origin to “explainable AI” literature that aims to make AI less like a “black box” [16, 42, 73]. As a result, when designing AI-powered DSTs, designers and researchers most often offered clinicians explanations of the AI's inner workings. The explanations include, for example, descriptions of training data, its machine learning model's training and prediction processes, and the model's

performance indicators [53, 54, 66, 74]. Practitioner-facing tools (e.g., Microsoft's HAX “responsible AI” toolkit, data sheets [24], model cards [46]) further boosted this approach's real-world impact. Experiments showed that AI explanations improved clinicians' satisfaction with DSTs and increased the likelihood of them taking the AI's advice [9, 42, 51, 60].

However, because AI suggestions are not always correct, “increasing the likelihood of clinicians taking the AI's advice” did not mean that clinicians made better decisions. To make better decisions, clinicians must calibrate their trust in each AI suggestion individually, on a case-by-case basis. Existing AI explanations largely failed in this regard. In clinical practice, the AI explanations were either unable to persuade clinicians to take the AI's correct advice (under-trust) [14, 26, 32, 36, 41, 73–75], or failed to enable clinicians to reject AI's errors (over-trust) [21, 32].

Recent research has identified several causes of AI explanations' repeated failure.

- **AI explanations are not always available.** Some clinical AI systems (such as deep learning systems for medical imaging [32]) are uninterpretable even to AI researchers, much more so to clinicians.
- **Too much information, too little time.** In clinical practice, clinicians make dozens, if not hundreds, of decisions in a day [19, 27], while each AI informs a decision. It can seem unrealistic to expect clinicians to comprehend all these AI's inner workings in addition to caring for patients [26, 34, 61, 70, 75].
- **AI explanations are not intuitive or actionable to clinicians.** In fast-paced clinical practice, clinicians need actionable information [13, 14]. Actionable information in the medical world typically means that the action has a causal relationship with desired patient outcomes, and ideally this causal relationship has been proven by randomized clinical trials on a large population [7, 30]. In contrast, AI models make personalized, data-driven predictions. The validity and actionability of AI suggestions, therefore, are counter-intuitive to many clinicians [75]. Studies showed that, even when clinicians understood how an AI works, they could not decide whether to take its suggestion. Instead, clinicians requested results from prospective randomized clinical trials that can show the causality between the AI features and patient outputs. They wanted these results to come from top medical journals [25, 34, 74, 75]; They wanted the information to be trustworthy inherently, rather than requiring them to validate it in the midst of their clinical decision-making [34]. For most clinical AI models, such evidence does not exist.

Recent research has started improving AI explanation designs to address these challenges. Some worked to make uninterpretable models more interpretable by appending purpose-built explanation-generation models to the DST [71], or offering counterfactual predictions demonstrating how changes to the values of AI features may impact their outputs [8, 38, 72]. Others addressed the “too much information, too little time” problem by presenting only selective or modular explanations to clinicians, the explanations that best justify the AI's suggestion [3, 17] or can best address clinicians' potential biases [35, 73]. These approaches are nascent and have not yet demonstrated an impact on clinicians' decision quality.

Interestingly, for few other researchers, the challenges listed above suggest that calibrating clinician-AI trust by explaining the AI is a “*false hope*” entirely; **clinicians need different kinds of information to validate AI suggestions** [16, 25, 26]. These researchers argued that AI explanations’ lack of actionability—one cannot decide whether an AI suggestion is right or wrong, even if one fully understands how the AI generated its suggestion—is a fundamental shortcoming that better explanation designs cannot fix. Instead of improving on the explanation designs, they promoted the idea that designers should take a step back and **work to understand clinicians’ information needs in validating AI suggestions, with considerations of their “identity, use contexts, social contexts, and environmental cues”** [35, 43]. This call for action is a key motivator for this research.

2.2 How Clinicians Validate Clinical Suggestions in Practice

To bootstrap our investigations into how clinicians validate care suggestions in their daily practice, we reviewed the literature on clinical hypothesis testing and decision-making. We noted three themes in this work: evidence-based best practices, the interpretive nature of many clinical decisions, and tool use.

Best practices: Evidence-Based Medicine (EBM). A culture of best practice dominates the medical world, therefore medical textbooks and best practices can offer some indications of clinicians’ actual practice. Known as Evidence-Based Medicine (EBM), clinical best practices require clinicians to use “*clinical evidence*” when examining patient-care-related suggestions and making decisions. “*Clinical evidence*” includes, for example, results of randomized controlled trials, standard care procedure concluded from multiple trials, the biological relationships between patients’ genome profiles and their treatment outcomes, and expert opinions on noteworthy case studies [11, 29, 31, 57, 63]. Biomedical literature is central to EBM best practices, because it documents all of these clinical evidence across medical domains [55, 65]. In medical schools, students learn to search for and user evidence in the literature to validate diagnostic or treatment hypotheses [30].

Medical textbooks instruct clinicians to, before adopting a piece of clinical evidence, examine both its *rigor* (How effectively did it establish the causal effect of the intervention and patient outcome?) and its *applicability* (Does the study result apply to their particular patient?) [64]. In examining rigor, clinicians should prioritize results from large-scale cohort studies over those from published case studies, according to the best practice known as the Level of Evidence Pyramid [48]. In examining applicability, clinicians should identify evidence that fits their patient’s population, the intervention under their consideration, intervention comparator, and the patient outcome of their interest, known as the PICO framework.

Empirical research showed that clinicians indeed use literature for decision-making in practice [20, 62]. When clinicians disagreed on a diagnosis or treatment, they often used literature as a shared source of truth [52, 59, 75]. The ways in which clinicians reacted to some AI explanations (e.g., requesting evidence of causality, see section 2.1) can seem reflective of these best practices.

Literature-based decision-support tools. When validating patient-care-related hypotheses or suggestions, clinicians most often used literature-based tools. These tools reflect both the EBM best practices and their need to rapidly mine seas of literature. For example, clinicians across disease areas use PubMed, a biomedical literature search engine [20, 49], which allows clinicians to search for clinical evidence based on their levels of rigor (e.g., randomized controlled trials, controlled trials, cohort studies, etc.) Many also used manually curated literature digest apps such as UpToDate [22] and Journal Club [1]. In addition, some hospitals hire in-house clinical librarians to assist clinicians in the literature search for decision-making [23, 37].

Recently, (semi-)automated literature digest tools have emerged, thanks to the rapid advances in biomedical Natural Language Processing and pre-trained language models such as GPT-3 and BERT [40]. For example, healthcare researchers have started building literature tools that search for clinical trials that match a patient situation based on the PICO framework [50, 58]. Such tools are available across multiple disease areas, such as cardiology, psychiatry, and COVID-19.

Social decision support and the interpretive nature of real-world clinical decisions. Notably, the situated and interpretative nature of many clinical decisions requires knowledge beyond what’s proven effective in past patient cases or clinical trials [61]. For example, for patients who arrived in Emergency Rooms while crushing and dying, clinicians often had to make decisions despite incomplete information, not to mention experimentally-proven evidence. In end-of-life treatment decision-making, the patient’s personal value is also often weighed vis-à-vis scientific evidence of treatment outcome [76]. In these decision contexts, clinicians resisted data-driven decision “*evidence*” provided by AI and preferred “*social decision support*” from colleagues: consulting each other and drawing from others’ tacit knowledge and experience [28, 75].

However, little empirical research has investigated how clinicians validated decision hypotheses (through literature, phone consultation, or others) and made care decisions as it naturally occurs. Instead, most work simply noted whether clinicians did or did not deviate from the EBM best practice [15, 33]. This line of research identified clinicians’ lack of time as one of the most significant barriers to practicing EBM at the point of care [15, 33].

3 METHOD

We wanted to explore new DST designs that can help clinicians validate its AI’s diagnoses and treatment suggestions on a case-by-case basis, convincing clinicians to take AI’s correct advice while rejecting its errors. Prior research has created many thoughtful DST designs by explaining how the AI worked. However, such explanations can seem vastly different from what clinical best practices instruct clinicians to seek when deliberating a care decision. Our work attempts to bring these strands of related work together. We first worked to understand how clinicians validated each other’s care suggestions in current practices (study 1). Building upon this empirical understanding, we then investigated how similar approaches might help clinicians validate AI’s advice (study 2).

Study 1	Clinical Domain	Experience	Study 2	Clinical Domain	Experience
P1	Neurology	10-30 yrs	P13	Internal Medicine	Over 30 yrs
P2	Cardiology	10-30 yrs	P14	Medical Student	2-5 yrs
P3	Pediatric Neurology	10-30 yrs	P15	Psychiatry	2-5 yrs
P4	Cardiology	10-30 yrs	P16	Psychiatry	1-2 yrs
P5	Neurogenetics	10-30 yrs	P17	Medical Informatics	1-2 yrs
P6	Nephrology	10-30 yrs	P18	Pediatric ER	Over 30 yrs
P7	Pathology	5-10 yrs	P19	Pharmacology	5-10 yrs
P8	Pediatric Nephrology	10-30 yrs	P20	Nursing	1-2 yrs
P9	Clinical Librarian	10-30 yrs	P21	Family Medicine	10-30 yrs
P10	ER	2-5 yrs			
P11	Clinical Librarian	10-30 yrs			
P12	Clinical Librarian	5-10 yrs			

Table 1: Study participants.

3.1 Stage 1: Investigating Natural Interactions

The first study aims to understand how clinicians examined each other's clinical suggestions in practice. Because biomedical literature is known to play a central role both in clinicians' everyday decision-making and their rationale for resisting AI suggestions [7, 30], we chose to focus on how clinicians sought, prioritized, and synthesized information from the literature to validate care suggestions. We paid particular attention to whether and how clinicians' behaviors deviated from best practices under the time pressure of patient care.

We conducted IRB-approved semi-structured interviews (with components of contextual inquiry) with 9 clinicians and 3 clinical librarians who these clinicians routinely hired to help with decision-making. We intentionally recruited participants from different clinical roles and specialties, ranging from nephrology physicians (P1, P6), pediatric physicians (P3, P8), cardiology physicians (P2, P4), a pediatric geneticist (P5), an emergency medicine medical intern (P10), and a pathologist (P7). We recruited initial participants from our collaborating hospitals and then expanded the set through snowball sampling. We conducted the interviews remotely. Each interview lasted for about 60 minutes.

In each interview, we started by inviting the participant to recall in detail a recent experience where they needed to examine a hypothesis or suggestion regarding patient care. We asked them to describe the specific patient situation and the broader clinical contexts. If and when they started describing their literature search, we invited them to share their screen and (re-)perform their search using the tools they used in actual practice (e.g., PubMed, Google Scholar). We invited them to think aloud, detailing how they sought, read, interpreted, and synthesized various information (within and beyond the literature) in relation to the specific patient decision and hypothesis in question. We asked follow-up questions to better understand the motivations that drive their information needs and advice-taking or -rejection behaviors.

We screen/video recorded and transcribed all interviews. We analyzed the data using a combination of affinity diagrams [47],

service blueprinting, and axial coding [5]. Through affinity diagramming, we analyzed (1) what information clinicians sought from the literature, (2) how they synthesized the information for the situated and interpretative patient situation at hand, and (3) how they prioritized such information when under time pressure. Further, service blueprinting allowed us to trace these information flow across the patient situation in question, clinicians' literature search, and clinicians' multiple decisions over the course of a patient's care. Next, we performed axial coding [12] to consolidate the perspectives of the clinicians who gave and received clinical suggestions. Finally, we confirmed our findings with four additional clinical professionals (a professor of biomedical informatics, a professor of medicine, and two orthopedic surgeons.)

3.2 Stage 2: Imitating Natural Interactions

After understanding how clinicians naturally validated each other's suggestions, we aimed to examine whether similar interactions can help clinicians validate AI suggestions. Towards this goal, we designed and prototyped a new form of DST that imitates clinicians' naturalistic interactions observed in study one. We then used the prototype as a Design Probe [44, 45], conducting IRB-approved interviews with clinicians for feedback and also iteratively improving the design based on their feedback.

We prototype three versions of the DST design, each focusing on one disease area (neurology, psychiatry, and palliative care.) To create a realistic user experience (UX), we populated the prototypes with three retrospective patient cases from top medical journals. We worked with two clinicians in selecting the cases and removed privacy-sensitive information from the clinical narratives. Finally, we populated the DST prototypes with AI diagnosis or treatment suggestions for these patients, using previously published, open-source ML models [68, 69].

To elicit honest feedback on the design, we recruited additional 9 clinicians who had not been interviewed earlier. We used the same recruitment process as in stage one. Study 2 participants come from Internal Medicine, Psychiatry, family medicine, emergency care, among other domains.

Decision Suggestions under Deliberation	Evidence Embedded in the Literature
Decide what information to inquire from a patient and actions to take during a patient visit;	Latest standard care procedures for particular symptoms; etiology;
Interpret lab test results and make diagnosis	The sensitivity and effectiveness of the lab test; differential diagnosis;
Make prognosis and treatment decisions	Different treatment options' anticipated outcomes based on outcomes of past patients similar to the patient situation in question (in terms of, for example, comorbidity).

Table 2: Clinicians share various literature evidence when they suggest different clinical actions to colleagues. The left column lists the clinical actions. The right column lists the information in the literature clinicians considered as “evidence” for that action.

In each interview, we first described the need for clinicians to deliberate whether to accept an AI-generated care suggestion. *Without showing our prototype*, we probed the participants' reactions to the idea of validating AI suggestions with literature. These reactions reflect clinicians' unbiased opinions regarding validating AI suggestions with literature. Next, we showed the participants a prototype that matched their medical expertise. We invited the participants to think aloud while they read the patient information, DST suggestions, and literature evidence. We asked follow-up questions to investigate how the literature evidence helped (or failed to help) them deliberate whether to accept the AI suggestions. Finally, we collected additional feedback by inviting clinicians to improve the prototype. We audio and video-recorded the interviews and analyzed the data using the same methods as Stage 1 (see 3.1).

4 STAGE ONE FINDINGS: HOW CLINICIANS NATURALLY VALIDATE SUGGESTIONS

In study one, our interviews confirmed that clinicians use biomedical literature to validate colleagues' care suggestions and to answer their “*tough questions*”, especially around rare patient cases. The literature is clinicians' shared source of truth. While all clinicians we interviewed said that they were “*very very busy*” and did not have time to read, almost all had looked up literature within the past two weeks, some even within the past few days.

Below we describe findings around the kinds of information clinicians used to deliberate care suggestions: (1) Clinicians provide *evidence* that can validate (or invalidate) their suggested diagnosis or treatment, rather than *explanations* of the suggestion; (2) Clinicians consider as a piece of “*perfect evidence*” as evidence that fits the patient situation at hand “*perfectly*”. Such evidence rarely exists; (3) When the “*perfect evidence*” does not exist, clinicians curate and synthesize imperfect evidence to make decisions; and (4) Clinicians utilize time-saving strategies to perform efficient reading, curation, and synthesis.

4.1 Don't Explain. Provide Evidence Instead.

When exchanging clinical suggestions, clinicians rarely explained how they came up with the suggestions (e.g., a suggested diagnosis, a lab test order, or a medication order). Instead, they shared a list of “*evidence*” from biomedical literature. Table 2 summarizes the clinical suggestions clinicians have referenced literature for and the information clinicians considered as “*evidence*” of the suggestions.

All clinicians we interviewed made an explicit effort to seek and share a list of evidence that is *scientific*. The resulting list is scientific in two ways. Firstly, it is comprehensive. Clinicians worked to ensure that they provided *all* supporting and opposing evidence of their suggestion in the literature. For example, when suggesting a treatment via email, one clinician shared “*the entire (literature) search results*” to demonstrate the evidence list is “*unbiased by ranking algorithms*” (P9). Others highlighted the latest literature on the list, signaling that they included “*all the latest data*” and treatment options (P1, P2). Secondly, the evidence list is reproducible. Clinicians often shared the link they used for their evidence search such that their colleagues can “*get the same results each time*” when examining the care suggestion and its list of evidence.

The comprehensive and reproducible evidence list served dual purposes in advice-giving-and-taking among clinicians. The list is persuasive. It demonstrated that the “*best evidence available*” supports the clinician's suggestion. On the other hand, the list allowed the clinician's colleagues to scrutinize the supporting and opposing evidence first-hand and draw their own conclusions.

4.2 The Myth of the Perfect Evidence

We use the term “*perfect evidence*” to refer to the kind of information that can immediately trigger clinicians' acceptance or rejection of a piece of clinical advice. In alignment with medical textbooks, clinicians characterized the perfect evidence as both rigorously generated (robustness) and highly applicable to the patient situation they face (applicability). However, interestingly, their criteria for evidence applicability far exceed what textbooks require.

When validating the rigor of a piece of literature evidence, clinicians used several proxies, such as the size of the patient population it has been tested on, publication venues, and authors. Some also checked the disclaimer part of the literature, assessing whether the clinical evidence on branded treatments was published for profit rather than in the patient's best interest (P2). Overall, the ways in which clinicians validated evidence rigor aligns with clinical best practices, though much faster. “*It's the reviewers' job, not mine, to invest in the time to see if the study is legit.*” (P8)

In validating the applicability of a piece of literature evidence, however, clinicians spent ample time and much cognitive energy. When reading clinical trial reports and case studies, almost all clinicians we interviewed jumped right to the first table in the article, which describes the demographic information of the patient or

patient population. They then worked on mapping the patient population onto their patients' situation. "[I] looked at what happened to these patients under what particular circumstances in the past, and sort of deciding whether they could apply to my patient." (P6) Some did this mentally. Others, especially the clinical librarians who routinely handle unusually complex patient cases, used the PICO (Population, Intervention, Comparator, Outcome) framework to match patient situations explicitly.

A pursuit of literature evidence that matches their patient's situation "perfectly" drives clinicians' search. Their definition of perfect applicability far exceeds what best practices demand (e.g., PICO match.) To all clinicians we interviewed, perfection means that the patient population in the clinical trial matches not only their patient's demographics but "all characteristics of the patient" (P4), including age, sex, weight, presence of comorbidities, past disease trajectories, previous interventions, and even family history. P8, a pediatric nephrologist, provided a particularly vivid example. When treating a preschool-aged female patient with Liddle syndrome and a family history of aldosterone-mediated hypertension, P8 suspected that the patient might have monogenic hypertension (a type of genetic high blood pressure.) He had run preliminary laboratory tests and looked up literature to decide what other tests he should order to confirm the diagnosis. He was looking for a testing procedure that has been proven effective on patients of the same gender and age, with the same disease variation (Liddle syndrome and monogenic hypertension), family history, and genetic inheritance pattern (e.g., the disease was similarly symptomatic in some family members). The evaluation of the suggested testing procedure "had to fit within those parameters" to convince him to carry out the suggestion immediately.

The perfect evidence does not exist for a vast majority of patient cases, especially for treatment suggestions. Combining the various criteria clinicians outlined above, the perfect evidence for the treatment of a patient is the treatment evaluation result from a "large-scale, well-conducted" randomized controlled trial, in which the trial participants are similar to the patient at hand in all characteristics. This is an exceptionally high bar, if at all achievable. Further, clinicians only seek help on unusual patient situations or rare diseases, making the perfect clinician-evidence-patient-situation match even less likely.

4.3 Making Decisions with Imperfect Evidence

Clinicians acknowledged that perfect evidence that can immediately trigger action is rare. After confirming that the perfect evidence does not exist for their patient, clinicians collected pieces of imperfect evidence; evidence that is useful but "non-conclusive" (P6). "You want to look for something else also." (P10)

Clinicians undertook a three-step process to identify and synthesize the list of imperfect evidence to conclude a care decision. First, clinicians collected a set of evidence from less "heavy hitting" journals. For example, they looked for insights from cohort studies, case-controlled studies, and case reports (rare patient cases that other clinicians encountered and published in peer-reviewed journals) (P4, P6, P10). When necessary, clinicians even referenced what they described as "obscure" (P3) or "gray literature" (P12), such as posters and conference meeting abstracts.

Next, if even less prestigious literature failed to reveal perfectly applicable evidence, clinicians relaxed their evidence applicability criteria. Using the PICO framework as a scaffold, clinicians first looked for studies or historical cases whose patient population and study setting overlap most with the patient situation they face. They referred to such evidence as "second-best matches". They then relaxed the search criteria further (for example, by including studies that cover a slightly different age group or a different subcategory of the same disease) until they had exhausted applicable evidence.

Finally, clinicians worked to identify a "trend" or "narrative" that can connect the pieces of evidence they collected from diverse sources with different levels of applicability. They again used the PICO framework as a scaffold. For example, to conclude a treatment decision, clinicians first "guesstimated" the likely treatment outcome of a patient population that shares some characteristics with their patient. They weighed competing evidence about the same population differently according to its robustness. Next, clinicians repeated this synthesis process on a patient population that shares another set of characteristics with their patient. This iterative process continues until they see a trend in the likely treatment outcome when these various patient characteristics meet. As P6 describes, "It's essentially looking at that and sort of deciding what happened to those patient [...] if they could apply to my patient." Clinicians took action on their patient according to this evidence trend.

It is not a coincidence that, when clinicians failed to find the perfect evidence, all of them relaxed the robustness criteria to look for the "second-best" patient-situation matches. When synthesizing evidence, clinicians used the PICO patient-situation match as a scaffold. Overall, the clinicians prioritized *applicability* over *robustness* when seeking and synthesizing evidence. They considered a highly applicable case study more informative than a well-conducted, large-scale randomized trial on a different patient population. P3's recollection of a recent experience punctuates the length clinicians would go to in order to harness a piece of less-robust but highly applicable information. When exploring treatments for a rare disease in a patient, P3, a neurologist, came across an ongoing clinical trial that fits the patient's demographics and history. He went on to contact a friend who works at the same hospital as the author (the clinician who was leading the trial), got introduced to the author, sought their "tips and tricks on how we are going to be successful with this [disease management]", went on to contact the drug company, and finally got the drug under trial for the patient. Clinicians value the evidence that fits their patient's situation well, in this case, even when the evidence is still under development.

"With Dr. [name], I wanted to know if he thought the drug would work. And then, if it did, how do I get it? How do I use it? What do I need to do? What resources do I need? How we are going to be successful with this?"

4.4 Time-Saving Strategies

The interview findings so far have shown that, when deciding whether to take a clinical suggestion, clinicians set an exceptionally high standard for decision evidence and thoughtfully curated and synthesized a comprehensive list of imperfect evidence when this standard is too high to reach. Noteworthy, clinicians also carried

out a set of time-saving strategies to make this process as efficient as possible.

Reading nothing but the evidence, its robustness, and its applicability. During the contextual inquiries, no clinicians read a clinical trial or case study entirely. Instead, they skipped to the population description to assess the study's applicability, then skipped to the Method section to validate its robustness. If the study met both criteria, clinicians skipped to results to find out whether the result *"was positive or not"* (P6). Following this process, it takes clinicians less than one minute to extract the evidence they need from an article of dozens of pages.

"The only thing that matters to me is that: No survival benefit with the addition of ADT (a medication). I won't read into the p-values and the percentages. All I care about is whether [outcome] is positive." (P10)

Clinicians also deemed literature with neutral outcomes as irrelevant because the information is not actionable. They stopped reading as soon as they realized the study outcome was neutral.

Synthesizing evidence (only) until it can justify action. While clinicians' literature synthesis for decision-making was comprehensive and systematic, the synthesis only went as far as it could justify a clinical action. For example, if the suggested clinical action is to order an additional lab test to confirm a potential diagnosis, clinicians only need a relatively weak set of evidence to take the suggestion because the cost of running the extra test is low, while the cost of missing a diagnosis is high. In this sense, clinicians inform patient decision-making with literature evidence much faster than a literature review process in research settings.

Reuse literature evidence over the course of a patient's care. Because clinicians sought literature evidence that matched the patient situation at hand, the same set of literature can inform multiple clinical decisions over the course of the patient's care (Table 2.) When first meeting the patients, clinicians accumulated *"landmark paper"* that can offer *"a rough guideline"* for care decisions but do not entirely match the patient's situation (P1, P6). As clinicians gathered more understanding about the patient situation (e.g., via diagnostic tests), they gradually turned to more precisely relevant clinical trials and even historic patient cases. The pieces of evidence they collected along the way progress from broad-stroke to case-specific, and all can potentially inform subsequent care decisions.

5 THE NEW DESIGN

We have so far described how clinicians examined each other's suggestions in current practice. Now we shift to consider how these naturalistic interactions inform how DSTs can help clinicians more effectively examine AI suggestions.

Recall that existing DSTs calibrated clinician-AI trust primarily by explaining to clinicians how the AI has generated its suggestions. Interestingly, clinicians used a very different type of information when deliberating on each other's clinical suggestions: biomedical literature. Specifically, we highlight four characteristics of literature information that made it effective.

- (1) **Providing evidence, rather than explanations.** Clinicians deliberated on colleagues' care suggestions, not by how the

colleague came up with the suggestion, but based on external evidence concluded from entirely different processes (e.g., clinical trials). Using this information, clinicians could stay focused on analyzing *"whether this is the right diagnosis or treatment for this patient"* rather than around *"whether the suggestion is trust-worthy"* by itself.

This providing-supporting/opposing-evidence approach can potentially address several challenges AI explanations face in calibrating clinician trust in AI suggestions. First, external evidence that can support or oppose a suggested diagnosis/treatment is always available, regardless of whether the AI models are interpretable or not. Second, it can save clinicians' time because clinicians are likely to be much more efficient in analyzing *"whether this is the right diagnosis or treatment for this patient"* than scrutinizing AI's inner workings. Finally, it can keep clinicians' cognitive work on clinical decision-making rather than AI-explanation sense-making.

- (2) **Providing both evidence of robustness and applicability, respectively.** Prior research has shown that clinicians did not consider explanations of how an AI generated its suggestions as actionable. The explanations often failed to sway clinicians to accept or reject the AI's suggestion [36, 41]. But why? Our findings revealed that a combination of two types of information could trigger clinicians' advice-taking behavior: evidence of the suggestion's *robustness* (also known as *"internal validity"* [64]) and of its *applicability* to the patient situation at hand (*"external validity"*). However, the perfect evidence that indicates both rarely exists. Therefore, clinicians assessed these two types of evidence separately: A study's population size indicates its robustness. Its patient population match indicates applicability. Existing DST designs did not always make this distinction. Instead, many DSTs unknowingly mixed evidence of its AI suggestion's *robustness* (e.g., evidence that the model is high performing on the patient population it is originally trained on) with evidence of its *applicability* (e.g., evidence that the model is high performing on a patient population that *"perfectly"* resembles the patient in question.) Further, evaluating a model's performance on *"patients like this patient in all characteristics"* is rare and challenging. In this light, it can seem unsurprising that AI performance indicators also struggled to persuade clinicians to accept or reject AI suggestions.
- (3) **Providing comprehensive rather than selective evidence.** A critical challenge in DST design is to calibrate clinicians' trust in AI under the time pressure of clinical practice. Some DST designs addressed this challenge by providing clinicians with selected information about the AI, such as a subset of features that can best justify the AI suggestion [26, 73]. Our interview findings suggest an alternative perspective to this approach. To clinicians, the trustworthiness of a clinical suggestion comes from the fact that the entirety of its supporting evidence, in aggregate, outweighs the entirety of its opposing evidence. Clinicians saved time by reading only the critical takeaway from every piece of evidence rather than reading particular pieces of evidence. DSTs might play a more critical role in calibrating clinician trust if it offered a comprehensive but succinct set of evidence and counter-evidence.

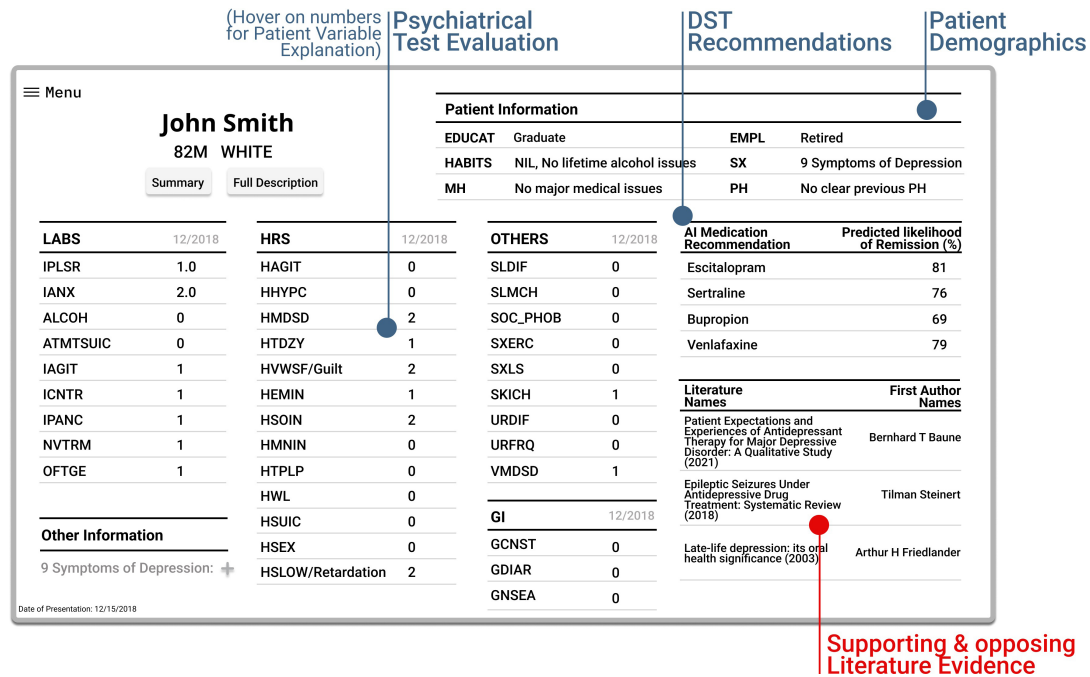


Figure 1: An initial prototype of the new DST design. This design helps clinicians deliberate whether to take an AI's suggestion by providing a list of supporting and opposing evidence from biomedical literature. The top panel highlights the patient's basic information (e.g., demographics). The columns on the left overview the patient's medical history and test results (in this example, psychiatric evaluation results). AI's treatment suggestion, along with its supporting and opposing evidence from biomedical literature, are on the right.

(4) **Drawing the evidence from a shared source of truth.** When a clinician shared a suggestion with a colleague, they used biomedical literature because it is a shared source of truth for both the advice-giver and the advice-taker. As a result, the clinicians did not need to elaborate on the trustworthiness of the literature evidence itself; they were both aware, for example, that randomized trials are more robust than historical case studies. Instead, they used the literature as a shared language and both focused on assessing the evidence's applicability to the specific patient situation at hand.

The same cannot be said about AI explanations. Explanations around AI's inner workings are the language of the AI (the advice-giver) rather than that of the clinicians (the advice-taker.) The struggles clinicians had in interpreting AI explanations can be seen as a symptom of this lacking-a-shared-source-of-truth problem (for example, they wanted evidence of causality between AI's features and patient outputs [74, 75]).

5.1 Design Goal and Strategies

We wanted to design a new form of DST that can help clinicians more effectively examine AI suggestions, such that they take only the correct suggestions and reject the AI errors. Embracing clinicians' naturalistic interactions, we identified the following design strategies:

- (1) Drawing the evidence from information sources that clinicians already understand and trust.

- (2) Providing evidence for/against the AI's suggestion, rather than explaining the AI's inner workings.
- (3) Providing evidence of the AI suggestion's rigor and applicability respectively.
- (4) Providing a comprehensive set of evidence and counter-evidence, but present each piece of evidence concisely.

5.2 The Design

Next, we translated the design strategies into a concrete interaction design and an initial prototype. Given the focus of this research, we focused on designing the part of the DST interface that calibrates clinicians' trust in AI suggestions (Figure 1, red arrow.)

- (1) **Drawing the evidence from information sources that clinicians already understand and trust.** We chose to use biomedical literature as the source of AI "explanation" information because clinicians across disease areas trust it as a source of decision evidence.
- (2) **Providing evidence for/against the AI's suggestion,** rather than explaining the AI's inner workings. Using the PICO framework, the new DST design retrieves biomedical articles that match the patient's situation (population) and the clinical action that the AI suggests taking (intervention, intervention comparator, and outcome of interest).
- (3) **Providing evidence of the AI suggestion's rigor and applicability respectively.** We included various types of literature, from randomized clinical trial reports (rigorous but imprecise

patient population match) to historical case studies (potentially precise patient situation matches, but less rigorous.)

- (4) **Provide a comprehensive set of evidence but present each piece of evidence concisely.** While presenting a comprehensive list of relevant articles, we summarized each in just a single sentence. Although drawn from study 1 observations, this design choice echoes existing counterfactual-based [8, 38] and modular AI explanation designs [3]. Noteworthy, our design differs in that its evidence comes from experimentally-proven facts rather than probabilistic AI predictions.

The design of the remainder of the DST interface simply followed the lessons from prior research [73, 75]. The resulting DST interface has three parts: Occupying most of the interface is the patient's demographic information, medical history, lab test results, and other EHR information. On the right side is AI's personalized diagnostics or treatment suggestion. Because we wanted to probe clinicians' feedback towards our design strategy "*providing evidence rather than AI explanation*", the prototype did not include any explanation of AI performance indicators or inner workings. This allowed us to observe whether they requested to use this information.

We populated the prototype with literature information using a combination of multiple functioning machine learning (ML) models and Wizard of Oz (WoZ, only for prediction quality control). First, we searched for relevant articles using a purpose-built search engine that is built upon EBM-NLP, a PICO-based biomedical literature dataset [50] and clinical bioBERT [40]. **We summarized each literature article by leveraging the extractive summarization function of GPT-3 (davinci-001), a pre-trained large language model [6].** Finally, we as "*wizards*" manually corrected the few errors in these ML-curated results (because the focus of this study is not to evaluate these ML models' document retrieval or summarization abilities. Instead, our goal is to probe whether our design goals and design choices are promising directions as such language models become increasingly capable.

6 STUDY TWO FINDINGS - HOW CLINICIANS VALIDATED AI SUGGESTIONS WITH LITERATURE

In study two, we interviewed an additional 9 clinicians for their feedback on the new DST design and collaboratively improved the prototype. Our findings confirmed that the design strategies drawn from clinicians' naturalistic suggestion-validation interactions can help calibrate their trust in AI suggestions. Moreover, clinicians' interactions with the prototype also revealed new design and research opportunities around (1) harnessing the complementary strengths of literature-based and predictive decision supports; (2) mitigating risks of de-skilling clinicians; and (3) offering low-data decision support with literature.

6.1 Confirming the Design Strategies

Literature evidence is a valuable complement to "big data". All clinicians we interviewed welcomed the shift of focus from explaining AI's inner workings to providing clinical evidence of the AI's suggestions. While the prototype included no explanations of the AI or its performance indicators, only one clinician requested some

explanation (She requested to see which hospitals' patient data the AI suggestions are trained on.) In contrast, clinicians described the causality-focused literature evidence as a valuable complement to the correlation-driven "*big data predictions*". They analogized AI predictions to "*aggregating many, many experiences of many doctors*" while describing literature evidence as embodying the mantra of Evidence-Based Medicine. The former resembles clinicians' tacit practical experience, while the latter offers proven knowledge and what "*most doctors decided was a good decision.*" Both cover critical aspects of clinical decision-making.

"There are different kinds of information that can inform your decision. [...] There's evidence-based medicine (EBM). [...] But experience is also very important. Mind you medicine (practice) is not really like mathematics. It is not an accurate science. It always has surprises and the common knowledge that people have from other doctors." (P21, family medicine physician)

Clinicians working in interdisciplinary medical domains (e.g., emergency care, family medicine, internal medicine) particularly appreciated the literature evidence. These domains cover "*a very, very large area of knowledge*". Literature evidence can help them examine a broader range of decision hypotheses and AI suggestions.

Clinicians highly valued the evidence of AI suggestion applicability. Echoing findings from study one, the prototyping study punctuated the critical importance of AI suggestions' applicability. Recall our initial prototype offered a brief summary of each biomedical article's findings. The clinicians we interviewed instead demanded an explicit description of the patient situation the article focuses on, even if it means more texts to read (See Figure 2 for the revised design). In order to more easily examine the literature evidence's applicability, some requested us to color code the different characteristics of the article's population of focus and to highlight the corresponding characteristics in the current patient's EHR information (P14, P15, P17). "*How the medicine fits into the patient's narratives is important.*" (P17).

"I need to figure out how to treat this really complex disease ... it's the treatment for one can kill the patient with the other". (P15, Emergency Room resident)

Clinicians demanded a concise summary of a comprehensive set of evidence. All clinicians we interviewed expected the literature evidence to be comprehensive. They considered comprehensiveness to be a key strength of literature. For example, while seeing "*big data*" as capturing patient outcomes of the lab tests and treatments that have long been available, clinicians considered literature as capturing the latest etiology, tests, and treatment options that can update their knowledge. The comprehensiveness of literature evidence is particularly valuable to clinicians when they discuss patient cases with colleagues and mentees.

"(I want to use this tool) when I'm presenting this [patient] case on the rounds to somebody, just put my thoughts together and make myself sound more on top of things. There is an element of showmanship in rounds, where you are trying to show how smart you are, that

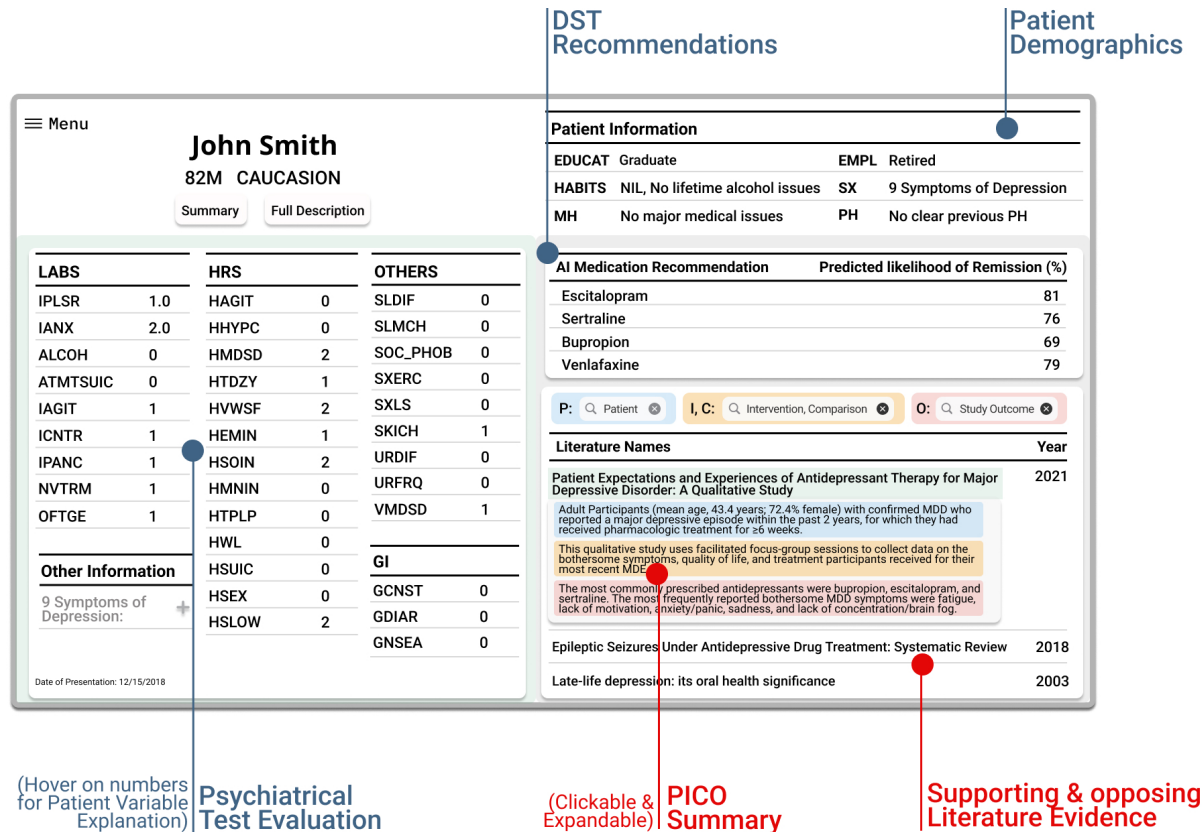


Figure 2: The revised DST design resulted from a co-prototyping study with 9 clinicians. The clinicians found the literature evidence useful in complimenting the data-driven DST predictions.

you’ve thought everything, that you got everything under control which you don’t.” (P13, Emergency Room physician)

Clinicians wanted the most concise summary possible while covering the literature evidence comprehensively. “Ideally, it can just give me some kind of alert”, an alert that the literature says what the AI suggests and what they are about to do are wrong. The Emergency Room clinicians we interviewed all requested the literature evidence to be summarized into such an alert, because otherwise “You know, if it’s 2 o’clock in the morning and I’m seeing a patient with phenomena, I’m not going to look at this.” Other clinicians expressed more appreciation of the detailed literature evidence, as they may study it before meeting patients or after work (e.g., “when you sit at home at night, thinking through this patient.”) In the meantime, they also wished for “literature evidence in a condensed form” such that they could use it when “sitting in front of patients and needing to decide something quickly”.

Clinicians trust biomedical literature for debasing AI suggestions. Our DST prototype intends to calibrate clinician-AI trust in diagnostic and treatment decision-making with literature evidence. Interestingly, clinicians appreciated the literature not only in these decision contexts, but also in debasing the AIs’ and their

own decision-making. This sentiment confirmed that to clinicians, literature is often a more trusted source of information than explanations of AI’s inner workings. It also suggests that DSTs can leverage literature for interactions beyond validating AI’s diagnostic and treatment suggestions.

“Literature can point out biases doctors made (in the AI training data), like black patients didn’t get (cardiac) catheterization as frequently as white patients did, and women didn’t get evaluated for coronary diseases as often as men did. Women present differently with coronary symptoms; they don’t describe [the symptoms] using the same words that were used in the textbooks, because the textbooks were based upon male patients. [The literature] It’s very thorough.” (P13, Emergency Room physician)

6.2 Revealing New Design Implications

The preceding observations confirmed that the types of information clinicians used to examine each other’s suggestions—external evidence of the suggestions’ rigor and applicability—could also help them examine AI suggestions. Noteworthy, biomedical literature and our prototype design represented merely only one source of

such external evidence and one way of presenting it. Clinicians' suggestions for improving the prototype revealed additional opportunities in this newly opened design space.

Harnessing additional sources of clinical evidence: medical images and -omics data. The clinicians we interviewed requested forms of clinical evidence beyond text-based literature summaries. For example, multiple physicians requested image-based literature evidence to help validate computer vision suggestions and make image-based care decisions. The “gold standard” reference images in pathology and dermatology literature can be valuable in these decision contexts. Similarly, pharmacists expressed a desire for literature evidence that matches patients’ “-omics data” (patient’s genetic or molecular profiles, which indicate their likely disease symptoms and medication responses).

Harnessing literature evidence as stand-alone decision support for low-resource hospitals. Multiple clinicians suggested that the literature evidence alone can be valuable for low-resource hospitals, such as rural health clinics and veterans’ hospitals. While some of these hospitals have only recently moved to EHR and may not have deployed AI-based DSTs, the literature evidence can nonetheless benefit their workforce, who are likely to be less experienced and in a more significant shortage of physicians.

6.3 De-skilling Clinicians? A Risk of Calibrating Clinician-AI Trust with Literature Evidence

Among the nine clinicians we interviewed, two (P13, P15) suggested a concern over clinicians becoming over-reliant on clinical literature. While other clinicians characterized the literature evidence as a trigger for self-reflection and slow-thinking, these clinicians analogized literature evidence to medical textbooks or even step-by-step instructions.

P15 describing her ideal form of literature evidence:
“It’s just kind of like someone made bullet points for other providers saying. Hey, this is what we do.”

In this context, they pushed back on seeing literature evidence on the DST, as it might over time undermine clinicians’ ‘ability to *‘think on their feet’* at the point of care. “A good physician should be able to practice medicine in a power failure.” P13 stated.

7 DISCUSSION

Clinical decision support tools (DSTs), powered by Artificial Intelligence (AI), promise to improve clinicians’ diagnosis and treatment decision-making process. However, no AI model is always correct. DSTs must enable clinicians to validate AI suggestions on a case-by-case basis, convincing them to take AI’s correct suggestions while rejecting its errors. Prior DST designs often explained AI’s inner workings or performance indicators. This paper provided an alternative perspective to this approach. Drawing from how clinicians validated each other’s suggestions in practice, we demonstrated that DSTs might become more effective in calibrating clinicians’ trust in AI suggestions if they provided evidence of the suggestions’ robustness and applicability to the specific patient situation at hand. Such evidence should be comprehensive but concise and come from a shared source of truth with clinicians.

This approach offers a timely answer to HCI communities’ call for a “*use-context-based*” approach to designing explainable AI [43]. Below, we first discuss how our findings point to new design opportunities around clinician-AI trust calibration and XAI. We then take a step back and more critically reflect on (2) the role literature-based AI and patient-history-based AI can and should play respectively in supporting clinician decision making.

7.1 Designing for Clinician-AI Trust Calibration

Like many others in the HCI community [25, 35, 75], we argue that “*explaining how AI work*” cannot fall calibrate clinicians’ trust in AI’s *individual* suggestions, even though it is indispensable in many other contexts (e.g., AI accountability, regulation). To help draw this distinction, we promote the terminology shift away from “XAI” as a catch-all term, to the concept of “*trust-calibration interaction design*”. In doing so, we hope to open up a new design space that explores a wide range of trust calibration information and interactions that go beyond explaining AI’s inner workings or performance indicators. The literature-based prototype presented in this paper provides one example. Future DST design practitioners should further explore this rich design space.

To jump start this exploration, our empirical findings identified three characteristics of biomedical literature that have made it effective in calibrating clinicians’ trust in AI. First, it contains information that is inherently trustworthy to clinicians, an essential requirement of clinical decision-support information [34]. Second, literature evidence is a *shared* source of truth among clinicians of all medical domains, therefore useful for real-world clinical decision making which is often highly collaborative and interdisciplinary. Third, unlike explanations of AI’s inner-workings, the literature contains various information that collectively can support all types of clinical decisions (e.g., etiology, diagnosis, treatment, medical imaging interpretation). When caring for each patient, clinicians make a series of interconnected decisions rather than discrete ones (e.g., identifying the cause of a symptom overlaps with concluding a diagnosis) [76]. Drawing from the same source of evidence across these decisions reduces clinicians’ cognitive load and saves time.

Future research should explore new information sources that share these characteristics (e.g., biomedical literature, medical image references, genomics data) to address AI’s potential errors and biases. Recent AI research that cross-checks EHR-based AI patient outcome predictions against the biological relations between treatment and outcome [2] and against clinician-authored Knowledge Graph [4] can be seen as great examples in this direction.

At a higher level, we advocate for an “*infrastructural approach*” to calibrate clinicians’ trust in clinical AI systems. Even though each EHR-based AI most often supports one clinician in making one decision a time, AI trust-calibration design should consider the broader context that clinicians collaborate constantly and make interconnected decisions. In this context, it is more effective for DSTs to draw from a consistent *source of information* that can validate diverse AI suggestions (regardless of whether it is a computer vision or a simple Bayesian model). DSTs can then adapt the *information presentation and interaction design* to each decision, its particular human context, and the AI system involved. Biomedical literature and genomics data have started becoming part of the healthcare

information infrastructure; Can DSTs harness this infrastructure and support clinicians' trust calibration in AI? What other information sources can become such a rising tide that raises all (AI) boats? These are exciting questions for future work to explore.

7.2 Blending Diverse Sources of Intelligence To Support Clinical Decision-Making

We have discussed the design and research opportunities that harnessing literature as a source of trust-calibration information has revealed. Parallel to these efforts should be work that critically reflects on the limitations of literature (or information that shares some of its characteristics) we have observed. We discuss these limitations against the two axes of difference that have scaffolded the debates around “*what constitutes the best evidence for clinical decision-making*”. These axes offer a useful structure for future research to investigate: How can DSTs best harness the complementary strengths of literature-based and EHR-based decision supports while ameliorating their weaknesses?

Evidence rigor and evidence applicability. The clinical world famously values causality: every clinical action ideally has a causal relationship with improved patient outcome, and the causality ideally has been proven by randomized clinical trials on a large, diverse patient population [7, 30]. Clinicians' focus on causality and large-scale evaluation is often described as causing their resistance to AI's highly-personalized decision suggestions [25, 34, 74, 75].

Our study 1 findings suggest that there is more to the story: Clinicians value the external validity of decision evidence (Does the information apply to this particular patient?) as much as its internal validity (How rigorously was it generated? Was it validated in a large patient population?) The perfect evidence is one that has been proven in a large patient population that resembles their patient's situation 100%. Such evidence simply does not exist. It is in this context that we do *not* see the success of our design as an indication that literature evidence should replace explanations of AI's inner workings. Instead, clinicians appreciated our design because it paired AI's personalized suggestion (high applicability) with causality-based literature evidence (high internal rigor). We see an opportunity for future research to reflect on and further improve the way our prototype coordinated AI explanations and literature evidence, so that they can together form the “*perfect*” evidence for clinicians' decisions.

Explicit knowledge and tacit knowledge. Evidence, whether it is from past patients' data or clinical literature, is not omnipotent. The situated and interpretative nature of many clinical decisions requires knowledge beyond what's proven effective in past patient cases or clinical trials [61]. But in what circumstances should one trust clinicians' “*experience*”, or tacit knowledge, and when should one instead force them to think slow and consider both the supporting and counter-evidence of their judgments? This is a question of long-standing debates in medicine and in AI decision support design [75].

This question also underlies the limitations of literature-based evidence. On the one hand, how can literature- and EHR-based AI systems enhance, rather than distract clinicians from their trained intuitions or even de-skill them? The emergency room physicians

we interviewed wished our literature evidence system took the form of an alert button “*Literature evidence in aggregate disagree with your judgment*”, pointing to an interesting direction for exploration. On the other hand, how can literature-based AI better serve decisions where clinicians' intuition falls short? For example, one clinician wished for a literature-based AI that reminds them of the gender and racial biases in old clinical trial results as well as in their own decision-making. By sharing these clinician critiques, we hope to start a reflective discussion about how HCI research can better harness the specific types of information in the literature to enhance clinician judgments in different ways.

8 CONCLUSION AND LIMITATIONS

This paper illustrates how clinicians used literature to validate each other's suggestions and presented a new DST that embraces such naturalistic interactions. The new design uses **GPT-3 to draw literature evidence that shows the AI suggestions' robustness and applicability (or the lack thereof)**. In doing so, we promote a new approach to “*explainable AI*” that focuses on not *explaining the AI* per se, but on designing information for intuitive AI trust calibration. In the midst of explosive growth in Foundational Language Models, this work revealed new design and research opportunities around (1) harnessing the complementary strengths of literature-based and predictive decision supports; (2) mitigating risks of de-skilling clinicians; and (3) offering low-data decision support with literature.

Parallel to work developing this new design approach, there should also be work critically examining it. Here, we also highlight two limitations of this work that merit further research. First, while the GPT-3-based prototype in this work was sufficient for most clinicians we interviewed, more work needs to understand the errors such models can make. Today's language technologies are by no means perfect in processing biomedical texts. Such errors can be particularly critical for Emergency Care, where clinicians concise summaries of literature evidence without losing necessary caveats and nuances. Second, this work studied clinicians from 14 different medical specialties. Future work should evaluate this literature-based design in more disease areas to further understand the scope of this generalizability.

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